FALI: Mechanistic Decomposition of Conditional Attention-Head Causal Usefulness

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Abstract

Mechanistic analyses of learned neural networks often compare component importance before and after an intervention, but detecting conditional importance does not explain why the component's causal contribution changed. We introduce FALI (Factorized Attention-Level Intervention), an output-grounded counterfactual methodology that decomposes the cross-state change in a selected attention head's causal usefulness into target, rival, and other attention-score contributions, value content, and surrounding-network context. Standard Owen attribution allocates interactions while reconciling the fine score decomposition with a coarse score/value/context game. Controlled synthetic systems and a randomly initialized transformer validate factor isolation, endpoint recovery, intervention isolation, and coarse–fine consistency. In GPT-2 Small on indirect object identification, L10H2 is positively recruited after primary Name Movers are ablated and exhibits a target-positive/rival-negative score decomposition, whereas L10H10 shows stronger target-attention adaptation without positive recruitment. A prospectively specified transfer test in Pythia-160M induction again finds positive recruitment, but the target and rival contributions reverse sign. Thus, the same high-level functional response can arise through different fine-grained mechanisms. FALI provides a verifiable mechanism-discrimination stage for selected conditionally important attention heads under explicit intervention semantics.

Keywords: mechanistic interpretability, causal intervention, transformer circuits, attention heads, conditional importance, causal attribution

1 Introduction

Machine-learning models can redistribute causal responsibility when their internal computation is perturbed. Ablating one component can change the contribution of others, producing self-repair, backup behavior, or more general conditional importance. For mechanistic interpretation, this creates a methodological problem: detecting that a component became more or less important does not identify why its contribution changed. A selected attention head may alter its task-relevant routing scores, its competing scores, or its value content; alternatively, essentially the same candidate computation may become more or less useful because the surrounding network has changed. Reliable mechanistic evidence therefore requires separating the detection of conditional importance from the explanation of its mechanism.

We introduce **FALI (Factorized Attention-Level Intervention)**, a hypothesis-driven methodology for resolving this ambiguity. FALI begins after a candidate attention head and a task-relevant target edge have been identified. It compares an intact state with a prespecified perturbed state and decomposes the resulting change in the candidate's *output-grounded causal usefulness*. The candidate's attention computation is factorized into target scores T, rival scores R, other structurally valid scores O, value content V, and surrounding-network context C. Evaluating the resulting 2^5 counterfactual states defines a causal game whose interactions are allocated with the standard Owen value. The fine score contributions reconcile exactly with a coarser SCORE/VALUE/CONTEXT decomposition.

The key distinction is between *localizing conditional importance* and *explaining it*. Activation and path patching identify behaviorally relevant components or paths; conditional co-ablation identifies components whose importance changes after a primary intervention; interaction-aware mediation analyses show that a component's measured effect can depend strongly on the rest of the model. FALI is downstream of these approaches: given a selected conditional change, it asks which architecture-aligned changes account for it under an explicit intervention model. This makes FALI a mechanism-resolution methodology rather than an automatic circuit-discovery procedure.

We first validate the intervention semantics in controlled systems. Deterministic synthetic attention models isolate each of T, R, O, V, C and a known cross-factor interaction, while a small randomly initialized transformer checks score-to-pattern reconstruction, value-to-output reconstruction, endpoint recovery, intervention isolation, batch-specific score partitions, and consistency between the fine Owen game and its coarse quotient game. These tests establish that the implementation recovers the factors it manipulates; they do not imply that the factorization is unique or that mixed hybrid states are naturally occurring model states.

We then apply FALI to GPT-2 Small on indirect object identification (IOI). Corrected functional analysis separates attention adaptation from recruitment: after the canonical primary Name Movers are removed, L10H2 becomes more causally useful, whereas L10H10 increases target attention even more strongly while its causal usefulness decreases. FALI assigns both heads a target-positive/rival-negative score pattern, showing that even a favorable-looking score reorganization does not itself

define recruitment. For positively recruited L10H2, the main fine-score contributions are $\phi_T > 0$ and $\phi_R < 0$ across the canonical confirmation cohort.

The decisive test is cross-setting transfer. We prospectively test the GPT-2 target-positive/rival-negative mechanism in Pythia-160M induction. Positive conditional recruitment appears again, but the fine score signs reverse: $\phi_T < 0$ and $\phi_R > 0$. The mechanism-transfer hypothesis therefore fails. The shared functional label—conditional recruitment—does not imply a shared fine-grained mechanism.

Our contributions are threefold. First, we define an output-grounded counterfactual estimand for the cross-state change in a selected attention head's causal usefulness. Second, we factor that change into score, value, and surrounding-context contributions, with target/rival/other resolution inside SCORE and coarse–fine consistency via standard Owen attribution. Third, we show empirically that two settings with positive conditional recruitment can exhibit opposite target/rival score mechanisms. The contribution is therefore a research methodology for discriminating mechanisms of conditional importance in learned neural networks, not a new theory of self-repair, a new localization method, or new Shapley/Owen mathematics.

An earlier public preprint from this project used the name FALI for a narrower **Frozen Attention Logit Intervention** analysis of attention-probability changes in GPT-2 IOI. The present paper substantially revises and supersedes that formulation: it changes the estimand to output-grounded causal usefulness, expands the intervention game to $T/R/O/V/C$, adds controlled implementation validation and coarse–fine reconciliation, broadens the GPT-2 confirmation, and adds a prospectively specified cross-setting mechanism-transfer test. The detailed relationship is stated in Related Work and in the MLJ Contribution Information Sheet.

2 Related Work

2.1 Self-repair and conditional importance

Mechanistic studies increasingly show that component importance is conditional on the state of the surrounding circuit. In GPT-2 IOI, primary and backup Name Mover heads provide a concrete setting in which multiple attention heads support the same behavior (K.R. Wang, Variengien, Conmy, Shlegeris, & Steinhardt, 2023). Subsequent work on the Hydra effect, copy suppression, and broader self-repair established that ablations can induce compensatory changes through multiple mechanisms rather than a single generic backup process (McDougall, Conmy, Rushing, McGrath, & Nanda, 2023; McGrath, Rahtz, Kramar, Mikulik, & Legg, 2023; Rushing & Nanda, 2024). Prompt-specific circuit analysis further cautions that even within a named task, mechanisms can vary across prompt families and model families (Franco, Tassis, Rohr, & Crovella, 2026).

FALI does not introduce self-repair or conditional importance. The closest operational precursor is Conditional Co-Ablation (CoAx), which measures how much a component's ablation effect changes after a primary set is removed and can recover backup components missed by first-order rankings (Gong, Zeng, Yuen, & Lim, 2026).

The distinction is one of explanatory stage: conditional ablation identifies *which* components become more important, whereas FALI takes a selected change in importance as the object to be explained.

2.2 Interaction-aware causal attribution

Activation patching and path patching test whether internal activations or paths causally influence behavior, but their conclusions depend on intervention choices and the state of the surrounding model (Goldowsky-Dill, MacLeod, Sato, & Arora, 2023; Heimersheim & Nanda, 2024; Zhang & Nanda, 2024). A particularly close conceptual neighbor is the causal-mediation analysis of Vaidyanathan, Arbour, Mueller, Niekum, and Jensen (2026), which shows that component effects can contain substantial interactions with the rest of the network. This rules out any claim that FALI is the first method to reveal context-dependent causal effects. FALI instead asks how a selected interaction-dependent change can be resolved into architecture-aligned factors of an attention head.

Related head-level work likewise finds that attention-head roles can depend on interactions with other heads (Nam et al., 2025). FALI is not a global head-role taxonomy or a learned gating method. It fixes one candidate head, two circuit states, and an explicit intervention algebra. Its context factor C therefore denotes the surrounding computation not explicitly supplied by the candidate score/value state; it can aggregate changes in other heads, MLPs, residual representations, normalization, and upstream or downstream computation. A large ϕ_C should not be read as localization of a unique external pathway.

2.3 Decomposing attention computation

Separating attention routing from transported content is an established distinction rather than a novel claim. Transformer-circuit analyses distinguish query–key (QK) computation from output/value (OV) computation, and later causal work further separates routing from value-mediated information flow (Elhage et al., 2021; Franco & Crovella, 2025; E. Wang, Wang, Bakalova, & Hahn, 2026). Factorial intervention studies also show that attention allocation and token content can be manipulated separately while remaining strongly coupled (Zhu, Zhang, Ding, & Yu, 2026).

FALI changes the quantity being decomposed. Its dependent variable is the *cross-state change in a selected head's output-grounded causal usefulness*, not an attention pattern or representation alone. SCORE and VALUE are therefore accompanied by an explicit CONTEXT factor; SCORE is further resolved into target, rival, and other valid score coordinates. Because these factors interact, we use the standard Owen value with the coalition structure SCORE=$\{T, R, O\}$, VALUE=$\{V\}$, CONTEXT=$\{C\}$ (Owen, 1977). The mathematics is established; the contribution is the intervention game and the architecture-aligned estimand to which it is applied.

Hybrid internal states may be off the model's natural activation manifold, so the resulting attributions are best interpreted as controlled counterfactual explanations under specified intervention semantics, not as evidence that the model implements five independent natural pathways (Grant, Han, Tartaglini, & Potts, 2026). This emphasis

on explicit intervention validation is consistent with broader efforts to evaluate mechanistic explanations against causal ground truth rather than narrative plausibility alone (Mueller et al., 2025).

An earlier public preprint used the name FALI for *Frozen Attention Logit Intervention*, a two-factor target-versus-background decomposition of attention-probability changes in GPT-2 IOI (Sharmila, 2026). The present work supersedes that interpretation. It changes the estimand to output-grounded causal usefulness, expands the game to $T/R/O/V/C$, introduces coarse–fine reconciliation through standard Owen attribution, validates the intervention semantics in controlled systems, and tests a prospectively specified cross-setting mechanism transfer. The historical two-factor formulation is retained only as a nested special case.

3 Problem Formulation

3.1 Conditional causal usefulness

Let f be a transformer evaluated by a scalar task metric m. We consider two executions of the same task instance: an intact state A and a perturbed state B, where B is produced by a prespecified intervention on a primary circuit component or set of components. Let h denote a candidate attention head whose change in functional importance we want to explain.

For context $s \in \{A, B\}$, let m_s^h be the task metric with h operating normally and m_s^0 the metric when h is zeroed while the rest of context s is held fixed. We define the candidate's output-grounded causal usefulness as

$$E_s(h) = m_s^h - m_s^0. \tag{1}$$

The cross-state change to be explained is

$$\Delta_h = E_B(h) - E_A(h). \tag{2}$$

We call Δ_h the *conditional gain*. Positive Δ_h means that the candidate's contribution becomes more positive after the primary intervention; it does not by itself imply that the candidate is helpful in the perturbed state. We use *positive functional recruitment* only when both $E_B(h) > 0$ and $\Delta_h > 0$.

Because the candidate-present and candidate-zero conditions are compared within the same surrounding context, Δ_h is not simply the change in task performance from A to B. It is the change in the candidate's marginal causal contribution under the specified intervention.

3.2 Mechanism-resolution objective

FALI begins after a candidate head and a task-relevant target edge or edge set have been specified, for example by prior circuit analysis, conditional co-ablation, patching, or a task-specific hypothesis. It does not search over heads or discover a circuit automatically.

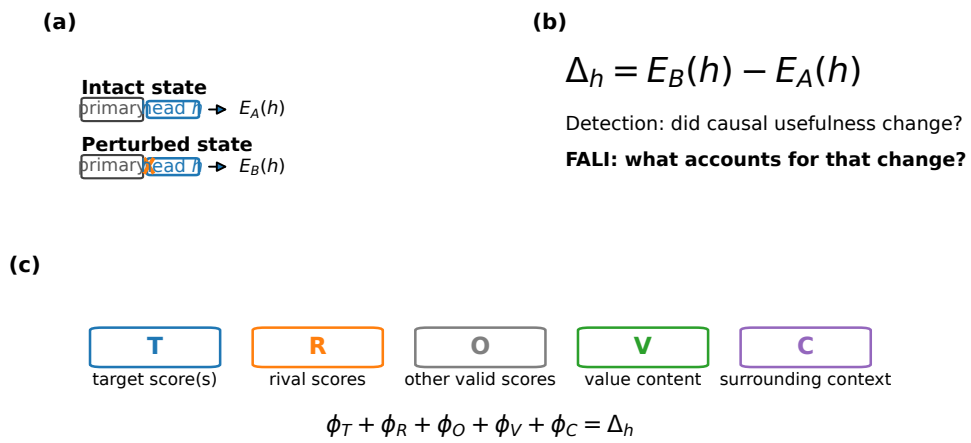


Fig. 1 FALI resolves a conditional change in a selected attention head's causal usefulness. (a) The same candidate head is evaluated in intact state A and prespecified perturbed state B. (b) The quantity to explain is $\Delta_h = E_B(h) - E_A(h)$. (c) FALI attributes this endpoint change across task-target scores T, rival scores R, other structurally valid scores O, value content V, and surrounding-network context C, with Owen efficiency $\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = \Delta_h$

The objective is to account for Δ_h using changes in three architecture-level objects: the candidate's attention scores, its value vectors, and the surrounding computation in which its output is used. We further partition the score tensor into the task-target score, competing scores in the same query row, and all other structurally valid scores. This yields a controlled mechanism-discrimination problem: under a fixed intervention algebra, which candidate-local or surrounding changes receive credit for the observed change in output-grounded usefulness? Figure 1 summarizes the setup.

4 Factorized Attention-Level Intervention

4.1 Architecture-aligned factors

For one candidate head and task instance, let S^A and S^B denote the cached pre-softmax attention-score tensors in the intact and perturbed executions, and let V^A and V^B denote the corresponding value tensors. For a right-padded causal sequence of valid length ℓ, we define the structurally valid score coordinates as

$$\mathcal{M} = \{(q, k) : q < \ell,\ k \le q\}. \tag{3}$$

Padding and future positions remain structural constants rather than intervention variables.

Let $q^\star$ be the task-relevant query and $k^\star$ the target key. We partition valid score coordinates into

$$T = \{(q^\star, k^\star)\}, \tag{4}$$

$$R = \{(q^\star, k) \in \mathcal{M} : k \neq k^\star\}, \tag{5}$$
$$O = \mathcal{M} \setminus (T \cup R). \tag{6}$$

Thus T contains the task-target score, R contains competing scores in the same query row, and O contains all other valid scores. The experiments in this paper use one target coordinate per example, although the definition extends to prespecified target sets.

FALI uses five binary factors, each taking its value from state A or state B:

- T: task-target score(s);
- R: rival scores in the task-relevant query row;
- O: all other structurally valid candidate-head scores;
- V: candidate value vectors;
- C: surrounding-network context.

The first three factors jointly determine the candidate attention pattern; V determines the content available to be transported; C selects the surrounding model execution outside the explicitly supplied candidate state. Importantly, C is a coarse residual context factor, not a localized downstream mechanism.

4.2 Counterfactual intervention game

For a cell $(\tau, \rho, \omega, \nu, c) \in \{0, 1\}^5$, we construct a hybrid score tensor by taking the T, R, and O coordinate blocks from their selected endpoints:

$$S_{\tau\rho\omega} = \mathbf{1}_T \odot S^\tau + \mathbf{1}_R \odot S^\rho + \mathbf{1}_O \odot S^\omega, \tag{7}$$

with invalid coordinates fixed by the structural mask. We recompute the attention pattern and candidate output as

$$P_{\tau\rho\omega} = \text{masked_softmax}(S_{\tau\rho\omega}), \tag{8}$$
$$z_{\tau\rho\omega\nu}[q] = \sum_k P_{\tau\rho\omega}[q, k] V^\nu[k]. \tag{9}$$

The reconstructed candidate output is injected at all valid query positions so that live candidate computation cannot leak into the hybrid state.

Let $F_c[z_h \leftarrow z]$ denote a forward pass in surrounding context c, with candidate head h overwritten by z. The utility of a FALI cell is

$$U(\tau, \rho, \omega, \nu, c) = m(F_c[z_h \leftarrow z_{\tau\rho\omega\nu}]) - m(F_c[z_h \leftarrow 0]). \tag{10}$$

The candidate-present and candidate-zero terms are always evaluated in the same surrounding context. The all-intact and all-perturbed cells recover the candidate effects in Eq. 1, so

$$U(1, 1, 1, 1, 1) - U(0, 0, 0, 0, 0) = \Delta_h. \tag{11}$$

FALI therefore defines a $2^5 = 32$ cell counterfactual game whose endpoint difference is exactly the change in candidate usefulness. Mixed cells need not be naturally occurring

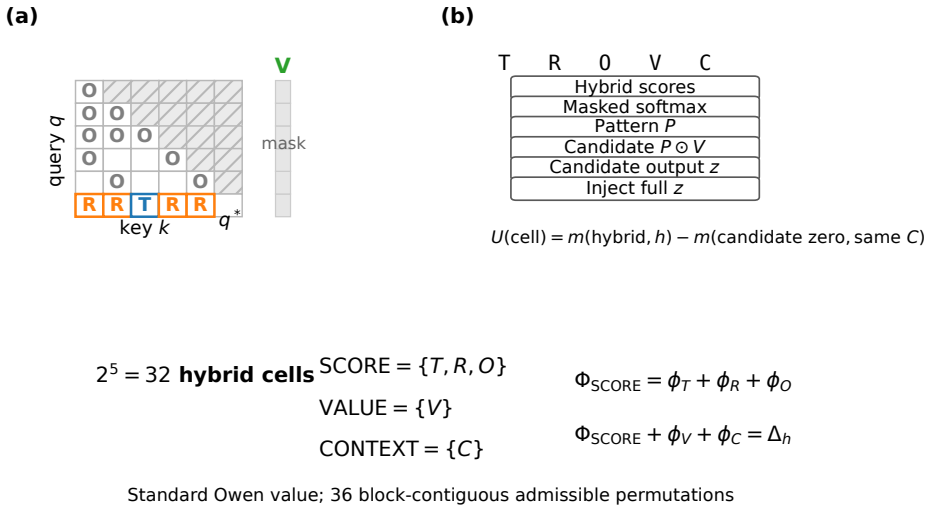


Fig. 2 FALI counterfactual intervention game. (a) Structurally valid pre-softmax scores are partitioned into task-target coordinate(s) T, rivals R in the same task-relevant query row, and other valid scores O; masked coordinates are excluded. (b) Each hybrid cell selects T, R, O, V, C from intact or perturbed endpoints, reconstructs the candidate attention computation, and measures utility relative to a candidate-zero run in the same context. (c) The $2^5 = 32$ cells are attributed with the standard Owen value under SCORE=$\{T, R, O\}$, VALUE=$\{V\}$, CONTEXT=$\{C\}$, yielding exact coarse–fine reconciliation

activation states; they are explicit interchange interventions. Figure 2 illustrates the construction.

4.3 Owen attribution and coarse–fine reconciliation

Let the players be $N = \{T, R, O, V, C\}$ and associate each coalition with the corresponding FALI hybrid. We use the standard Owen value with the fixed coalition structure

$$\Pi = \{\text{SCORE} = \{T, R, O\},\ \text{VALUE} = \{V\},\ \text{CONTEXT} = \{C\}\}. \tag{12}$$

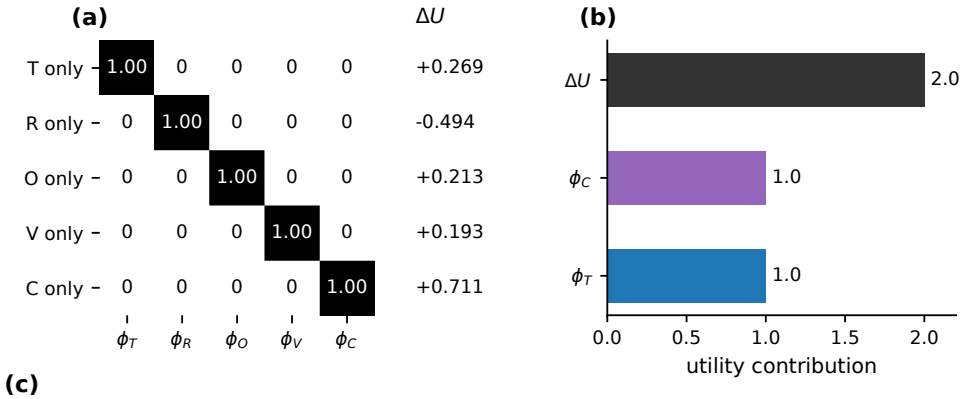
This averages marginal contributions over all block-respecting permutations rather than imposing a fixed sequential ordering (Owen, 1977). The full expression and equivalent 36-permutation implementation are given in the Supplementary Information.

Two identities are central. First, efficiency allocates the complete endpoint change:

$$\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = \Delta_h. \tag{13}$$

Second, the fine score attributions reconcile with the three-player quotient game:

$$\Phi_{\text{SCORE}} = \phi_T + \phi_R + \phi_O, \tag{14}$$



(c)

Score to pattern reconstruction	0
Pattern/value to candidate output	0
Mask and batch-specific partitions	PASS
Independent endpoint recovery	0
Candidate isolation / live leakage	0
Coarse-fine quotient discrepancy	1.11×10^{-16}

Overall validation: PASS

Fig. 3 Controlled validation of the FALI intervention semantics. (a) In five deterministic single-factor systems, attribution is isolated to the manipulated factor; the values at right give the corresponding endpoint changes. (b) A pure $T \times C$ interaction with total change approximately 2 receives the expected 1/1 Owen split between T and C. (c) Random-transformer reconstruction, endpoint, intervention-isolation, and quotient-game checks pass, with maximum coarse–fine discrepancy 1.11×10^{-16}

while ϕ_V and ϕ_C equal the VALUE and CONTEXT quotient attributions. These values allocate interaction under the specified game; they are not pure direct effects and need not correspond to unique natural pathways.

The historical two-factor attention-probability formulation is nested as a special case: if the utility is changed to target attention probability, O, V, C are fixed, and only T and R vary, the game reduces to an ordinary two-player Shapley decomposition. The present framework changes both the intervention scope and, crucially, the quantity being explained.

5 Method Validation

Before applying FALI to pretrained models, we tested whether the intervention game recovers known factors in controlled systems and reproduces ordinary transformer computation at its natural endpoints. These checks validate implementation semantics, not learned-model mechanisms. Figure 3 summarizes the main results; exact values and the validation chronology are provided in the Supplementary Information.

5.1 Synthetic ground truth

We constructed deterministic attention systems containing masked softmax, values, $z = PV$, and a context-specific readout, then evaluated all 32 FALI cells. In five single-factor systems, only the manipulated factor received attribution up to floating-point precision. A separate pure $T \times C$ interaction produced the analytically expected equal Owen allocation between T and C, with the remaining factors at zero. The all-intact and all-perturbed hybrids reconstructed their endpoints exactly at recorded precision, and collapsing T, R, O into SCORE reproduced the corresponding SCORE/VALUE/CONTEXT quotient game with maximum discrepancy 1.11×10^{-16}.

5.2 Transformer implementation invariants

We then repeated the mechanical checks in a randomly initialized two-layer TransformerLens model. Masked softmax reconstructed the cached attention patterns, pattern–value multiplication reconstructed the candidate head output, batch-specific $T/R/O$ masks formed a complete non-overlapping partition of valid scores, and candidate-output injection prevented live candidate computation from leaking into hybrid cells.

Independent ordinary forward passes gave intact effect 0.455046, conditional effect -0.085684, and conditional gain -0.540729; the corresponding FALI endpoints reproduced all three with zero recorded discrepancy. Non-candidate heads remained unchanged under candidate-output overwrite, and the fine game agreed exactly with the collapsed coarse quotient game.

Together, these tests establish factor isolation, structural masking, endpoint recovery, intervention isolation, and coarse–fine reconciliation for the implemented pipeline. They do not establish that the factorization is unique, that hybrid states lie on the model's natural activation manifold, or that a large attribution identifies a unique natural pathway.

Use of generative AI tools.

Generative AI tools, including OpenAI ChatGPT and Cursor, were used during manuscript development for literature-search support, drafting and editing, code generation and review, figure-layout development, and LaTeX/repository integration. All experimental designs, analyses, numerical results, citations, scientific claims, and final manuscript content were reviewed and verified by the authors against the underlying source code, result artifacts, and primary literature. The authors retain full responsibility for the work and its conclusions.

6 GPT-2 IOI: Recruitment and Mechanism

6.1 Does attention increase imply recruitment?

We apply FALI to GPT-2 Small on indirect object identification (IOI), perturbing the canonical primary Name Movers L9H6, L9H9, and L10H0 and using final-position IO-minus-subject logit difference as the task metric (K.R. Wang et al., 2023). The canonical confirmation cohort contains 15 low-level templates crossed with

BABA/ABBA structure, 30 prompts per stratum, and four seeds, for 3600 prompts total. L10H2 and L10H10 are the primary candidate heads; L9H7 and L9H0 serve as same-layer architectural controls. Full sampling and bootstrap details are given in the Supplementary Information.

L10H2 satisfies the positive-recruitment criterion on the broad canonical cohort. Its conditional effect is 0.382 (95% CI $[0.341, 0.424]$) and its conditional gain is 0.298 ($[0.263, 0.335]$). Its target attention also increases by 0.279 ($[0.262, 0.296]$).

L10H10 shows why attention adaptation is not sufficient evidence of recruitment. Its target attention increases even more, by 0.310 ($[0.285, 0.335]$), yet its conditional gain is negative, -0.263 ($[-0.302, -0.225]$), and its conditional effect is also negative, -0.079 ($[-0.111, -0.046]$). Figure 4a therefore separates a visible attention response from the output-grounded functional response that FALI is designed to explain.

6.2 What changes account for the GPT-2 response?

For positively recruited L10H2, the coarse SCORE contribution is positive ($\Phi_{\mathrm{SCORE}} = 0.445$), while value contributes approximately zero and surrounding context contributes negatively ($\phi_C = -0.146$). Within SCORE, the dominant pattern is target-positive and rival-negative: $\phi_T = 0.783$ (95% CI $[0.722, 0.838]$) and $\phi_R = -0.341$ ($[-0.372, -0.311]$), with $\phi_O = 0.0039$. The directional pattern holds in every seed and every canonical stratum, and remains when the legacy-overlapping template is excluded.

L10H10 exhibits the same fine-score signs despite its negative functional gain: $\phi_T = 0.329$ ($[0.300, 0.355]$) and $\phi_R = -0.126$ ($[-0.139, -0.113]$). Its positive SCORE contribution (0.206) is outweighed by a substantially negative surrounding-context contribution ($\phi_C = -0.471$). Figure 4b–c shows both the fine score decomposition and the coarse score/value/context accounting.

This comparison is important for interpretation. A target-positive/rival-negative score reorganization is not synonymous with recruitment: it can occur in a head whose net causal usefulness decreases. Recruitment is defined by the output-grounded candidate effect; FALI then explains how the cross-state change is allocated under the intervention game. The same-layer controls L9H7 and L9H0 provide the complementary architectural case: their candidate score/value states do not change across the primary intervention, so their measured cross-state changes are assigned to surrounding context up to numerical precision.

7 Robustness and Representation Sensitivity

7.1 Alternative primary interventions

We tested whether the GPT-2 target-positive/rival-negative pattern persisted under matched-mean and matched-resample alternatives to the canonical zero ablation. It did. For L10H2, matched mean gives $\phi_T = 0.440$ and $\phi_R = -0.166$, while matched resample gives $\phi_T = 0.265$ and $\phi_R = -0.0474$. L10H10 retains the same signs under

both alternatives. Hierarchical confidence intervals exclude zero in the predicted direction, and the sign pattern is preserved across all canonical strata. Full estimates are reported in the Supplementary Information.

This channel-level robustness should not be confused with an overall robustness pass. The preregistered experiment-level criterion also required a valid zero-ablation damage reference; that reference failed, with estimate -0.319 and 95% CI $[-0.396, -0.244]$. We therefore conclude only that the fine-channel directions persist under the tested matched alternatives. We do not claim robustness to arbitrary ablation schemes.

7.2 Softmax-gauge sensitivity

Pre-softmax attention scores are defined only up to a per-query additive constant under softmax. Fine attributions obtained by intervening directly on raw scores can therefore depend on the chosen logit representation even when the resulting attention probabilities are unchanged. Native-logit FALI remains our primary analysis; we additionally evaluate a prospectively specified sensitivity in which valid scores are row-centered before hybridization.

The qualitative fine-channel conclusions survive this gauge fix. In GPT-2 IOI, L10H2 retains $\phi_T > 0$ and $\phi_R < 0$ (native $0.783, -0.341$; row-centered $0.644, -0.202$), and L10H10 likewise retains $T > 0, R < 0$ (native $0.329, -0.126$; row-centered $0.306, -0.103$). In Pythia induction, the reversed pattern also remains: L8H2 retains $T < 0, R > 0$ (native $-0.144, +0.118$; row-centered $-0.047, +0.021$). Coarse SCORE, value, context, and endpoint conditional gain remain unchanged at recorded precision, while the fine magnitudes shift, especially in Pythia.

The supported conclusion is deliberately narrow: *the fine-channel sign conclusions are robust to the prespecified zero-mean gauge sensitivity analysis.* This does not establish complete gauge invariance, magnitude invariance, or a uniquely privileged score representation. Detailed paired results are provided in the Supplementary Information.

8 Cross-Setting Transfer: Pythia Induction

8.1 Can the GPT-2 fine mechanism transfer?

We test the GPT-2 target-positive/rival-negative pattern in a different model family and task: Pythia-160M on repeated-token induction. The transfer hypothesis was specified prospectively: after selecting a later head that becomes more useful when an earlier induction head is removed, FALI should again yield $\phi_T > 0$ and $\phi_R < 0$.

Head selection was separated from FALI attribution. Discovery used seed 42 only and did not inspect fine FALI attributions; the frozen selection procedure chose L5H0 as the primary head and L8H2 as the candidate. Confirmation used independent seeds 1, 8, and 22 across offsets 8, 16, and 24, with 128 examples per offset per seed. The repeated-token construction, matched control, sampling details, and bootstrap procedure are described in the Supplementary Information.

8.2 Recruitment transfers; the fine mechanism does not

The held-out cohorts satisfy the functional prerequisites. The induction advantage is 14.017 (95% CI $[12.280, 15.939]$), removing L5H0 produces a primary drop of 1.146 ($[0.887, 1.539]$), and L8H2 is positively recruited: conditional effect 0.621 ($[0.456, 0.860]$) and conditional gain 0.486 ($[0.326, 0.722]$). Figure 5a therefore establishes the shared high-level phenomenon: positive conditional recruitment occurs in both the GPT-2 IOI and Pythia induction cases studied here.

The fine mechanism, however, reverses. For Pythia L8H2, FALI gives $\phi_T = -0.144$ (95% CI $[-0.247, -0.084]$) and $\phi_R = +0.118$ ($[0.058, 0.219]$), with the reversed signs in every confirmation seed and offset. The other score and value channels are small ($\phi_O = -0.00027$, $\phi_V = -0.00210$), while the surrounding-context contribution is positive and large ($\phi_C = 0.514$). As elsewhere, C denotes coarse surrounding computation rather than a uniquely localized mechanism.

Figure 5b places the two settings side by side: GPT-2 IOI recruitment is target-positive and rival-negative, whereas Pythia induction recruitment is target-negative and rival-positive. The preregistered mechanism-transfer hypothesis therefore fails. No alternate head pair, model, task, or threshold was substituted after observing this result.

The inference is intentionally bounded. The experiment shows that positive conditional recruitment can occur with opposite fine target/rival organizations under the specified FALI games. It does not establish that recruitment mechanisms are universally different across model families, nor that no deeper common mechanism exists. Because model, task, head pair, perturbation, and metric all change between the two settings, the experiment is a mechanism-discrimination test rather than a controlled explanation of which factor caused the reversal.

9 Discussion

9.1 A shared functional response can conceal different mechanisms

The central empirical result is a failed mechanism-transfer hypothesis. GPT-2 IOI L10H2 becomes more useful after removal of the canonical primary Name Movers and shows target-positive/rival-negative score contributions. Pythia-160M induction L8H2 also becomes more useful after its primary intervention, yet the fine score signs reverse. The two settings therefore share the functional label *conditional recruitment* without sharing the tested fine-grained mechanism.

This distinction matters because recruitment is defined by a change in causal usefulness, not by a particular internal adjustment. GPT-2 L10H10 makes the same point within a single model and task: it increases target attention and exhibits the same target-positive/rival-negative score organization as recruited L10H2, yet its overall conditional gain is negative. Neither an attention increase nor the sign of one FALI channel is sufficient to define recruitment. The output-grounded candidate effect defines the functional response; FALI then allocates the cross-state change under the specified intervention game.

The cross-setting reversal should not be over-interpreted. Model family, task, head pair, perturbation, and metric all differ between GPT-2 IOI and Pythia induction. The experiment therefore shows that the GPT-2 fine-score pattern is not a universal consequence of positive conditional recruitment in the tested settings. It does not identify which of those differences caused the reversal, nor does it rule out deeper common structure.

9.2 Where FALI fits in the interpretability workflow

FALI is best viewed as a mechanism-resolution stage downstream of localization. Activation and path patching identify behaviorally relevant activations or paths (Goldowsky-Dill et al., 2023; Heimersheim & Nanda, 2024; Zhang & Nanda, 2024); conditional methods such as CoAx identify components whose importance changes after a primary intervention (Gong et al., 2026); interaction-aware mediation analyses establish that component effects can depend on the surrounding network (Vaidyanathan et al., 2026). FALI begins once a selected attention head and target edge are already specified, and asks which score, value, or surrounding-context changes account for its altered usefulness.

The framework also does not claim novelty for QK/OV separation or Owen attribution. Routing-versus-content distinctions are established in transformer-circuit analysis and more recent causal decompositions (Elhage et al., 2021; Franco & Crovella, 2025; E. Wang et al., 2026). Standard Owen values provide interaction-aware accounting under a fixed coalition structure. FALI's contribution is to apply these ingredients to a different estimand: the intervention-induced change in one selected head's output-grounded causal usefulness, with target/rival/other resolution inside SCORE and an explicit surrounding-context factor.

This framing suggests a practical pipeline: first localize a component or detect conditional importance; then, when the scientific question is *why that selected attention head's usefulness changed*, apply FALI under a prospectively specified intervention game. The output should be treated as a discriminative counterfactual explanation, not as an automatically discovered natural pathway.

10 Limitations

Hypothesis-driven scope.

FALI requires a selected attention head and a task-relevant target edge or edge set. IOI and repeated-token induction provide clear target coordinates; less structured tasks may require substantial prior circuit analysis. Because the five-factor game evaluates 32 counterfactual cells per example and candidate, FALI is intended for focused mechanism resolution rather than exhaustive whole-model scanning. Candidate or target selection after inspecting FALI results would introduce substantial researcher degrees of freedom, so confirmatory use should freeze these choices prospectively.

Counterfactual hybrids.

Mixed $T/R/O/V/C$ states need not occur during natural model execution. They answer what the model would do under the specified interchange intervention, not

which latent state it would otherwise visit. Representation-level interventions can move computation away from naturally occurring internal distributions (Grant et al., 2026); FALI attributions therefore do not identify five independent natural pathways.

Intervention and coalition dependence.

Attributions depend on the intact and perturbed endpoints, the candidate-zero baseline, the score partition, and the SCORE/VALUE/CONTEXT coalition structure. Owen attribution allocates interactions consistently within that game, but a different defensible intervention or grouping can allocate credit differently. In GPT-2, the target-positive/rival-negative pattern persists under the tested matched alternatives (mean and resample), while the broader preregistered robustness criterion fails because the zero-ablation reference is invalid. The evidence therefore supports only the intervention variants actually tested.

Score representation dependence.

Fine $T/R/O$ attributions operate on pre-softmax scores, which admit per-row additive gauge freedom. The primary native-logit conclusions retain their signs under a prospectively specified zero-mean row-centering sensitivity, but the magnitudes change, particularly in Pythia. We therefore claim sign robustness to that sensitivity, not general gauge invariance or magnitude invariance.

Coarse context factor.

C aggregates surrounding computation outside the explicitly supplied candidate score/value state, potentially including other heads, MLPs, residual representations, normalization, and upstream or downstream changes. A large ϕ_C assigns credit to this block of the game but does not identify the external component or pathway responsible. Resolving C would require additional hypotheses and interventions.

Empirical scope.

The substantive evidence covers GPT-2 Small IOI and Pythia-160M repeated-token induction. This is sufficient to falsify transfer of the preregistered GPT-2 target/rival pattern to the tested Pythia setting, but not to characterize recruitment mechanisms across architectures, scales, tasks, or training stages. Prompt-specific circuit results reinforce this distributional caution (Franco et al., 2026). The present factorization is also attention-head-specific; MLPs, residual features, or distributed mechanisms would require different architecture-aligned factors rather than mechanically reusing $T/R/O/V/C$.

11 Conclusion

FALI explains how a selected attention head's causal usefulness changes after a circuit perturbation by separating target, rival, and other score changes from value content and surrounding-network context. The framework is output-grounded, interaction-aware, and explicit about the counterfactual intervention semantics under which its attributions are defined.

The experiments show why this resolution is useful. In GPT-2 IOI, attention adaptation is not sufficient for positive recruitment, and a target-positive/rival-negative score pattern can occur even when a head becomes less useful. More importantly, positive conditional recruitment appears in both GPT-2 IOI and Pythia induction while their target/rival score contributions have opposite signs. The preregistered mechanism-transfer hypothesis therefore fails.

The resulting claim is deliberately bounded: a common functional label need not imply a common fine-grained mechanism. FALI provides a way to test that distinction for selected attention heads rather than infer mechanism from conditional importance or attention change alone.

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Competing Interests

The authors declare that they have no financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

Data and Code Availability

The study uses publicly available pretrained models and programmatically generated experimental datasets. Processed datasets, aggregate result files, preregistration records, and analysis artifacts supporting the reported results will be archived with the accompanying code release. The code required to reproduce the reported analyses will be made publicly available in the project repository and archived in a persistent release.

Author Contributions

Ronith Sharmila: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization, Project administration. Delen Dursun: Supervision, Writing – review and editing. Rupesh Agrawal: Supervision, Writing – review and editing. All authors read and approved the final manuscript.

References

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., ... Olah, C. (2021). A mathematical framework for transformer circuits. *Transformer Circuits Thread*, , Retrieved from https://transformer-circuits.pub/2021/framework/index.html

Franco, G., & Crovella, M. (2025). Pinpointing attention-causal communication in language models. *Advances in neural information processing systems* (Vol. 38).

Franco, G., Tassis, L.M., Rohr, A., Crovella, M. (2026). Finding highly interpretable prompt-specific circuits in language models. *arXiv preprint arXiv:2602.13483*, , Retrieved from https://arxiv.org/abs/2602.13483

Goldowsky-Dill, N., MacLeod, C., Sato, L., Arora, A. (2023). Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*, , Retrieved from https://arxiv.org/abs/2304.05969

Gong, Z., Zeng, Z., Yuen, C., Lim, W.Y.B. (2026). Conditional co-ablation: Recovering self-repair backups in transformer circuits. *arXiv preprint arXiv:2607.01940*, , Retrieved from https://arxiv.org/abs/2607.01940

Grant, S., Han, S.J., Tartaglini, A., Potts, C. (2026). Addressing divergent representations from causal interventions on neural networks. *International conference on learning representations.* (ICLR 2026 proceedings)

Heimersheim, S., & Nanda, N. (2024). How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, , Retrieved from https://arxiv.org/abs/2404.15255

McDougall, C., Conmy, A., Rushing, C., McGrath, T., Nanda, N. (2023). Copy suppression: Comprehensively understanding an attention head. *arXiv preprint arXiv:2310.04625*, , Retrieved from https://arxiv.org/abs/2310.04625

McGrath, T., Rahtz, M., Kramar, J., Mikulik, V., Legg, S. (2023). The hydra effect: Emergent self-repair in language model computations. *arXiv preprint arXiv:2307.15771*, , Retrieved from https://arxiv.org/abs/2307.15771

Mueller, A., Geiger, A., Wiegreffe, S., Arad, D., Arcuschin, I., Belfki, A., ... Belinkov, Y. (2025). MIB: A mechanistic interpretability benchmark. *Proceedings of the*

42nd international conference on machine learning (Vol. 267, pp. 45069–45108). Retrieved from https://proceedings.mlr.press/v267/mueller25a.html

Nam, A., Conklin, H., Yang, Y., Griffiths, T., Cohen, J., Leslie, S.-J. (2025). Causal head gating: A framework for interpreting roles of attention heads in transformers. *Advances in neural information processing systems.* Retrieved from https://arxiv.org/abs/2505.13737

Owen, G. (1977). Values of games with a priori unions. *Mathematical economics and game theory: Essays in honor of oskar morgenstern.* New York: Springer.

Rushing, C., & Nanda, N. (2024). Explorations of self-repair in language models. *arXiv preprint arXiv:2402.15390*, , Retrieved from https://arxiv.org/abs/2402.15390

Sharmila, R. (2026, May). *When attention repairs itself: A mechanistic study of backup heads in gpt-2.* Zenodo. Retrieved from https://doi.org/10.5281/zenodo.20350000

Vaidyanathan, S., Arbour, D., Mueller, A., Niekum, S., Jensen, D. (2026). The curse of multiple mediators: Hidden interaction effects in activation patching. *arXiv preprint arXiv:2606.27510*, , Retrieved from https://arxiv.org/abs/2606.27510

Wang, E., Wang, Y., Bakalova, A., Hahn, M. (2026). How few-shot examples add up: A causal decomposition of function vectors in in-context learning. *International conference on machine learning.* Retrieved from https://arxiv.org/abs/2605.16591

Wang, K.R., Variengien, A., Conmy, A., Shlegeris, B., Steinhardt, J. (2023). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *International conference on learning representations.* Retrieved from https://arxiv.org/abs/2211.00593

Zhang, F., & Nanda, N. (2024). Towards best practices of activation patching in language models: Metrics and methods. *International conference on learning representations.* Retrieved from https://arxiv.org/abs/2309.16042

Zhu, J., Zhang, Y., Ding, J., Yu, X. (2026). Lexical perturbations disrupt LLM reasoning: An empirical study of attention diversion. *arXiv preprint arXiv:2608.22140*, , Retrieved from https://arxiv.org/abs/2608.22140

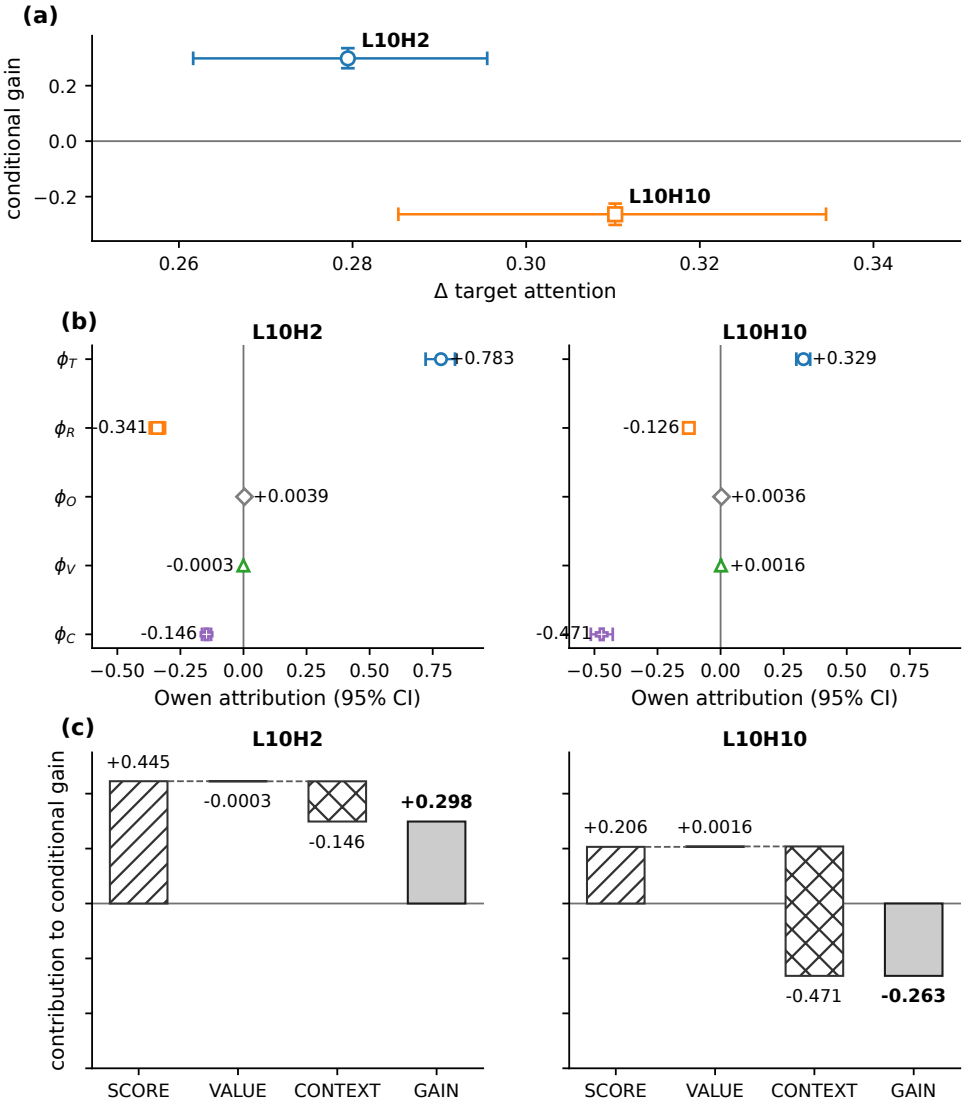


Fig. 4 GPT-2 IOI separates attention adaptation from positive functional recruitment. (a) L10H2 and L10H10 both increase attention to the indirect-object target after ablation of the canonical primary Name Movers, but their conditional gains have opposite signs. (b) Both heads show positive target-score and negative rival-score FALI contributions, while O and V remain near zero on the plotted scale. (c) Coarse score/value/context accounting explains the difference: L10H10 retains a positive SCORE contribution, but its negative surrounding-context contribution outweighs it and yields negative conditional gain. Points are balanced means; intervals are hierarchical 95% bootstrap confidence intervals

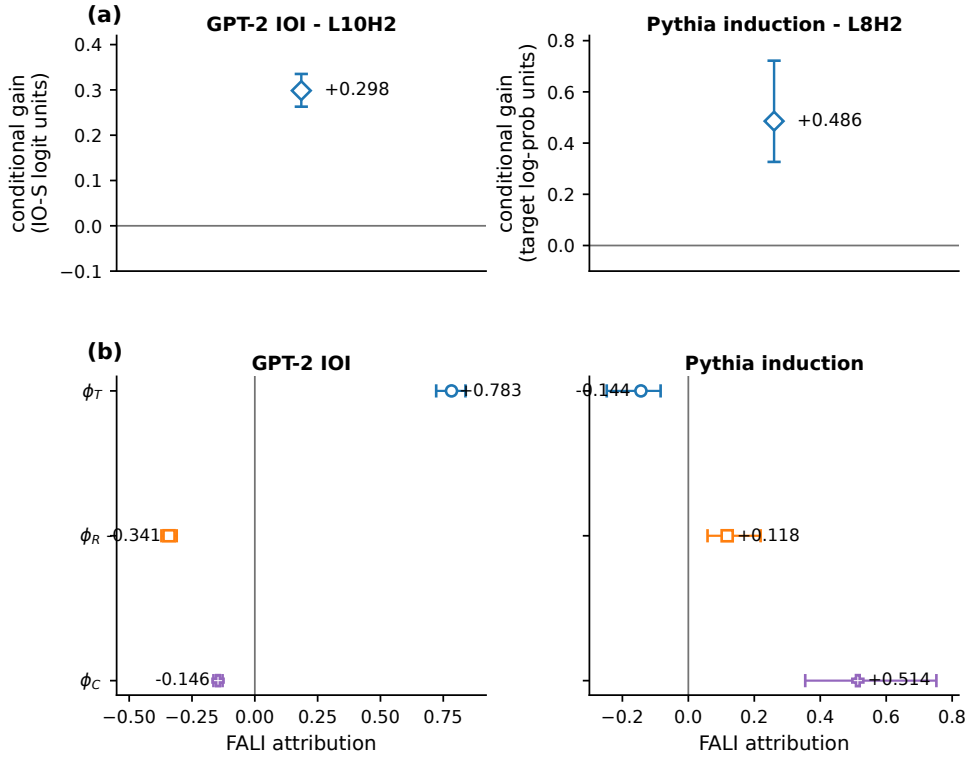


Fig. 5 Conditional recruitment occurs in both studied settings, but the preregistered fine-score mechanism does not transfer. (a) GPT-2 IOI L10H2 and Pythia-160M induction L8H2 both have positive conditional gain after their respective primary interventions; task-specific axes are used because the utility metrics differ. (b) The fine attribution signs reverse: GPT-2 is target-positive/rival-negative, whereas Pythia is target-negative/rival-positive; their surrounding-context contributions also differ in sign. Points are balanced means and intervals are the prespecified 95% bootstrap confidence intervals for each experiment